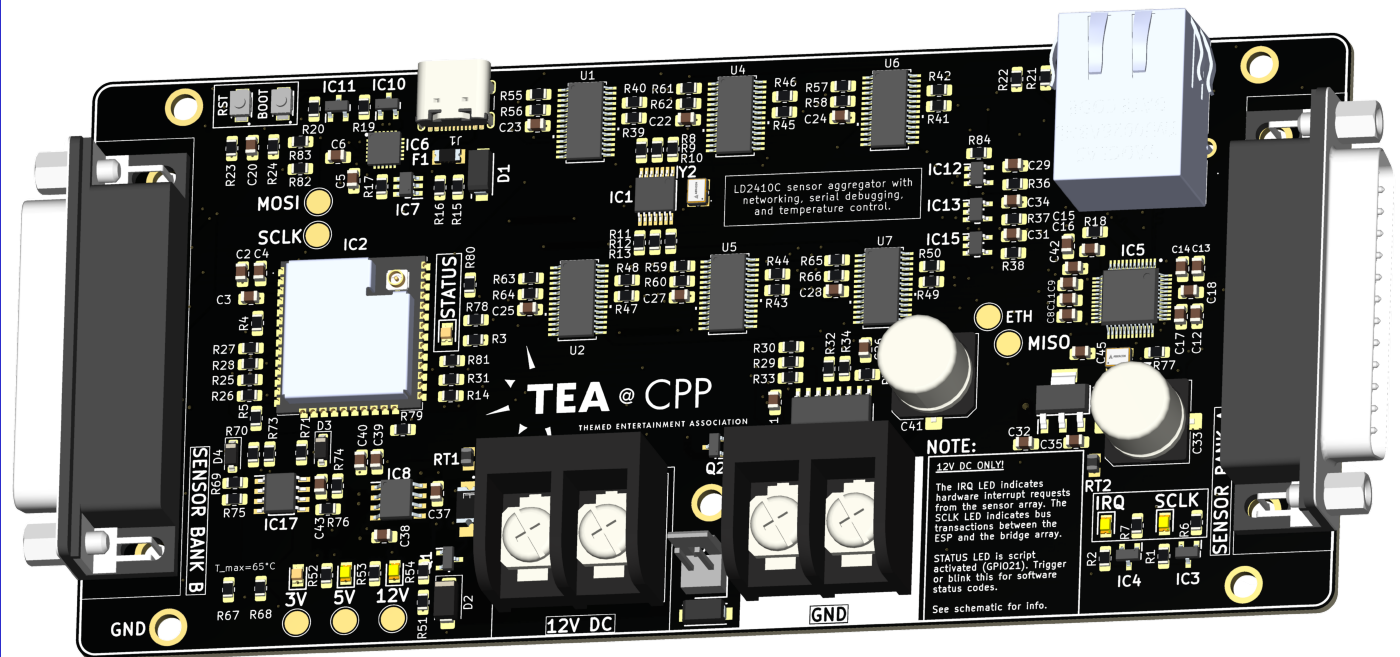


TEA@CPP Escape Room 2026: "The Holy Grail"

An industrial-grade radar sensor aggregator board that bridges 12x LD2410C 24GHz sensors to an ESP32 via a buffered UART-to-SPI network. Features dual-stage voltage regulation, bus isolation, temperature control, and dedicated serial and networking ports.

Major Components:

ESP32-WROOM-32UE
SC16IS752IPW_128
CD74HC138QM96Q1
SN74LVC1G125DBVT
5PB1106PGGI8
7447715003
LMR33640
AP63205WU-7
5747842-2
W5500
USBL6-2SC6
TMJ0026ABNL
MMBT3904
CP2102N-A02-GQFN24
1812L150_24MR
LM393DR



Designers:
William Dinkins
Arriana Flores

Contact: whbcool1998@gmail.com



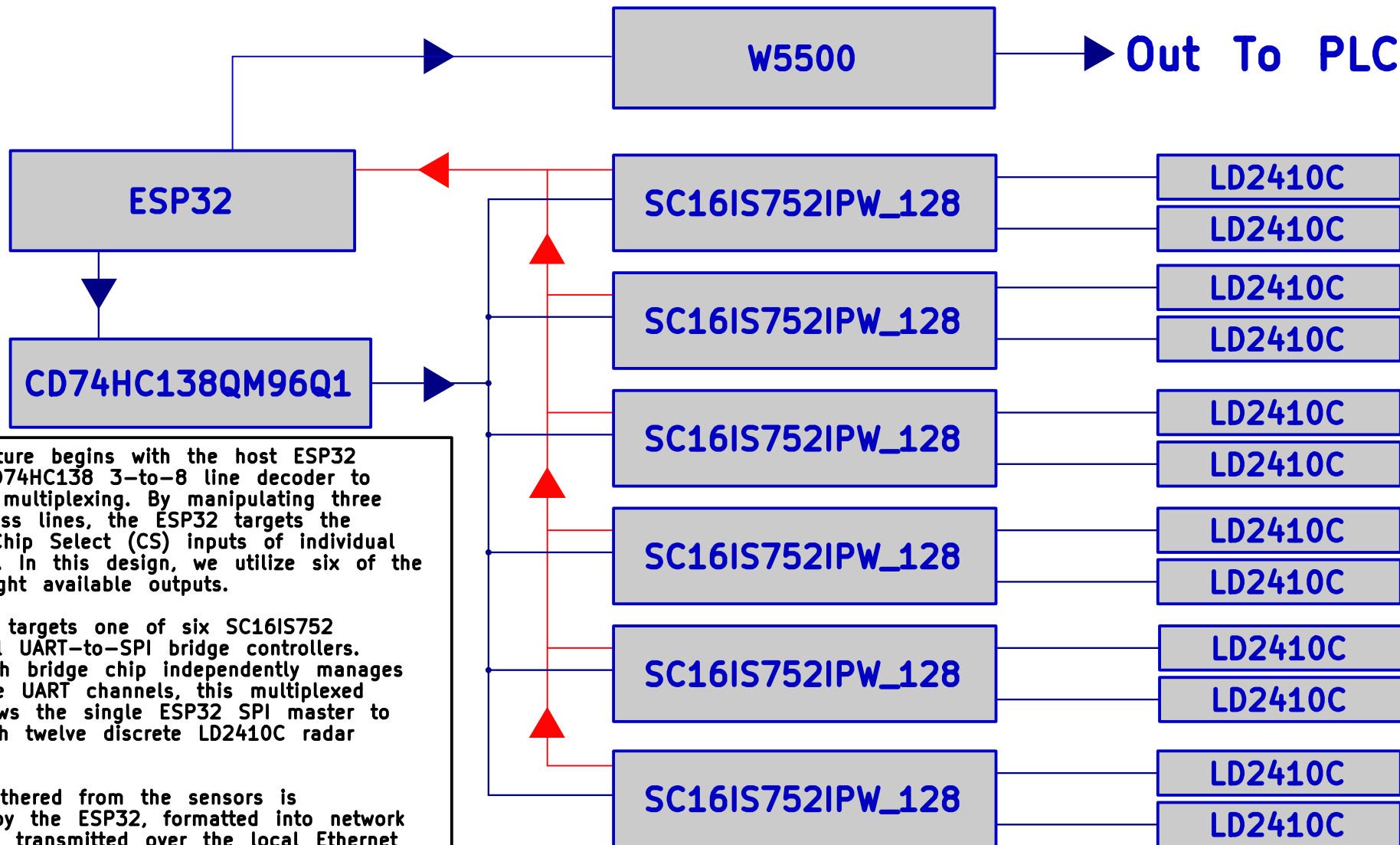
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Title: The Holy Grail

Size: A4 Date: 2026-04-19

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The architecture begins with the host ESP32 driving a CD74HC138 3-to-8 line decoder to control bus multiplexing. By manipulating three digital address lines, the ESP32 targets the active-low Chip Select (CS) inputs of individual bridge chips. In this design, we utilize six of the decoder's eight available outputs.

Each output targets one of six SC16IS752 dual-channel UART-to-SPI bridge controllers. Because each bridge chip independently manages two hardware UART channels, this multiplexed network allows the single ESP32 SPI master to interface with twelve discrete LD2410C radar sensors.

Telemetry gathered from the sensors is aggregated by the ESP32, formatted into network packets, and transmitted over the local Ethernet show network.

NOTE: The original target of this board was to read 24 sensors on one board. To achieve 24 sensors, this board should be duplicated and ran in parallel. Adding 2 more SC16IS752 chips to this design increases the max theoretical capacity to 16 sensors, but requires a complete board redesign.

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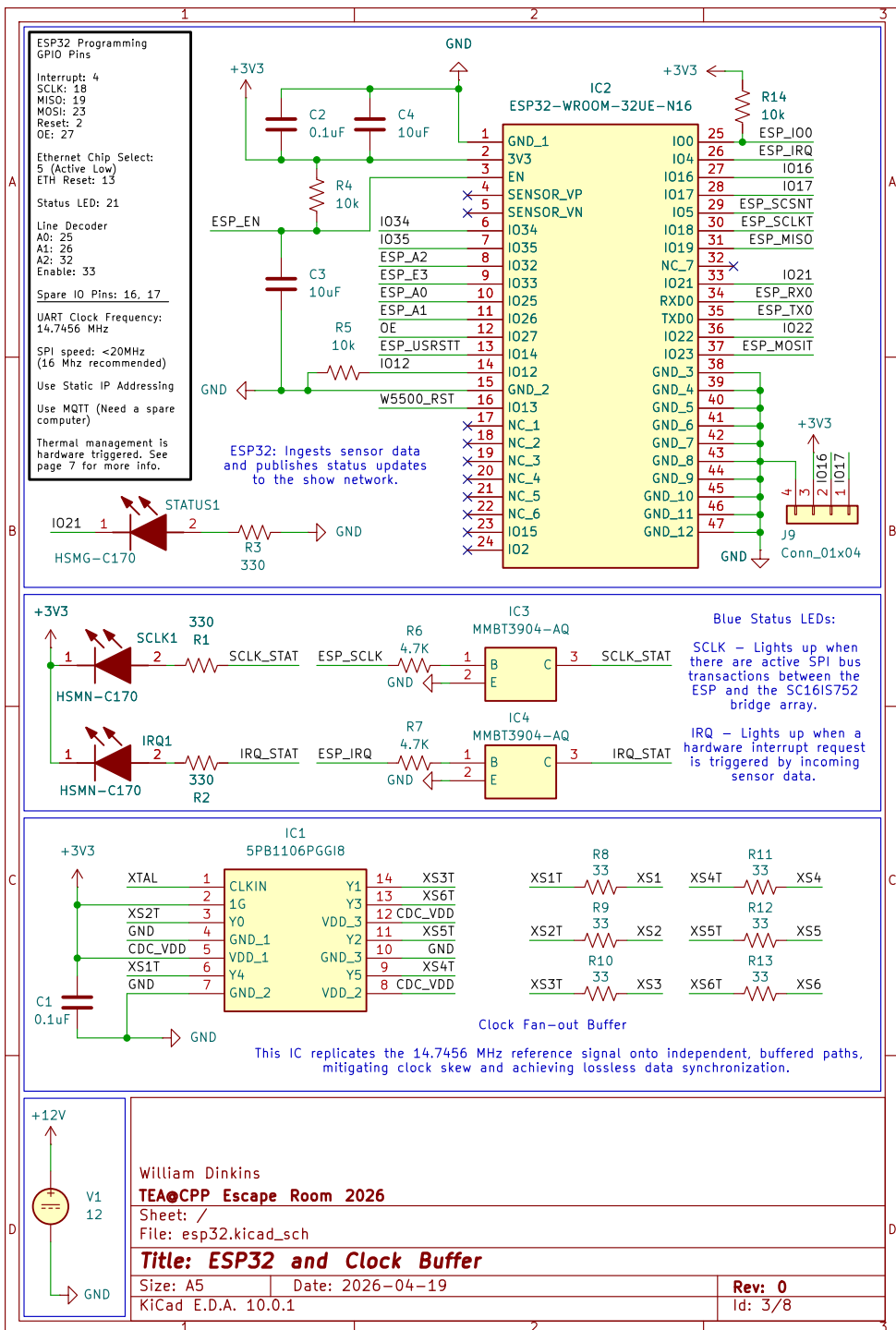
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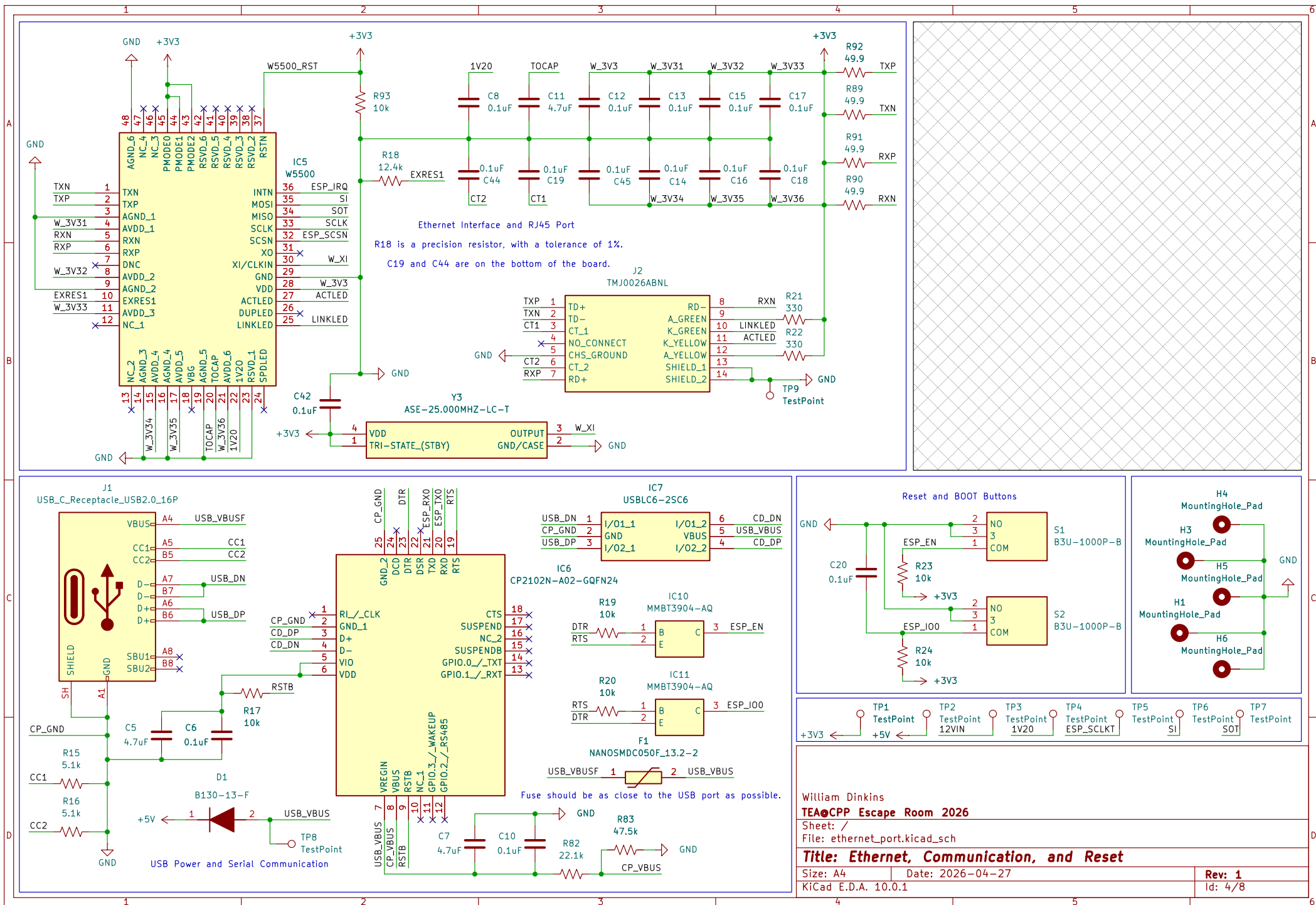
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Size: A4 Date: 2026-04-27

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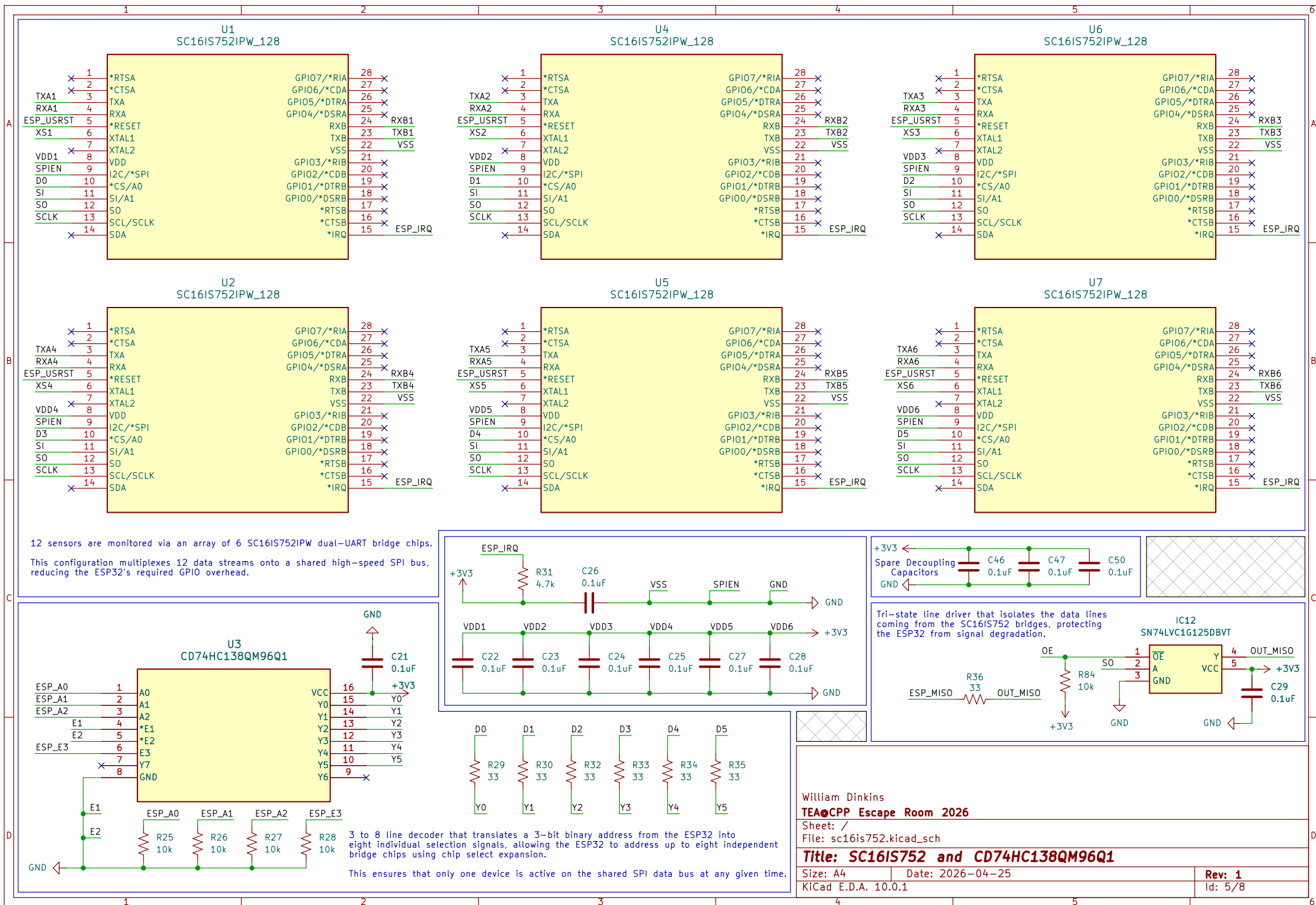
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Title: Ethernet, Communication, and Reset

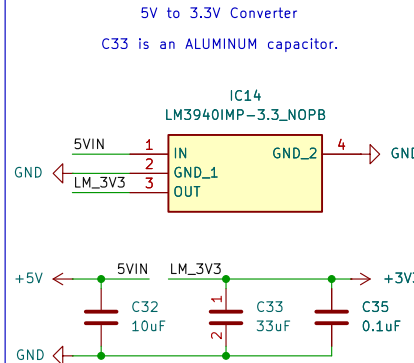
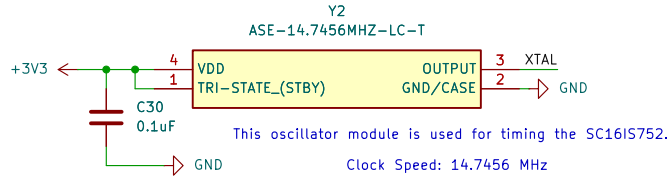
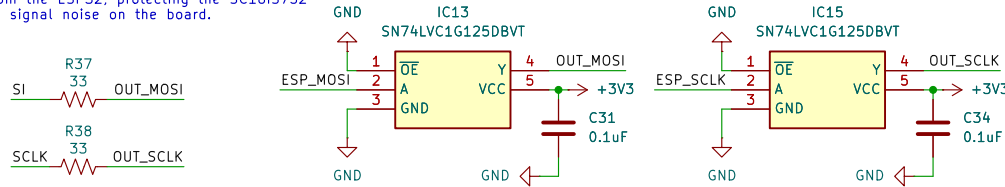
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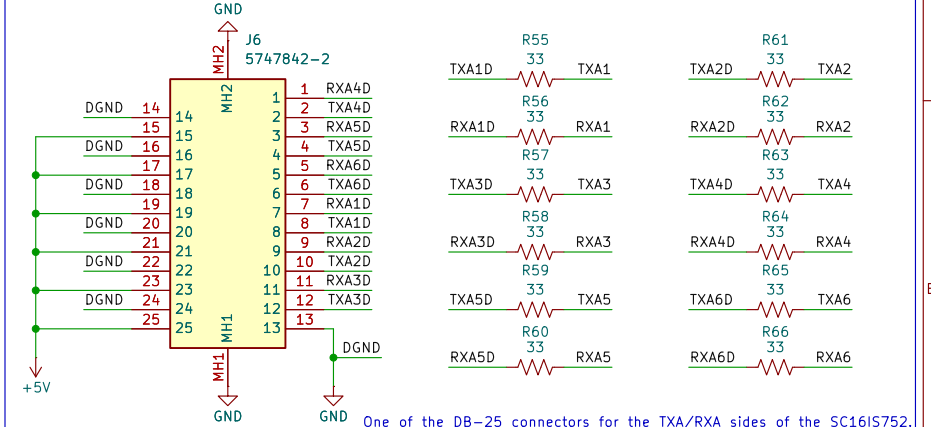
Tri-state line driver that isolates the data lines coming from the ESP32, protecting the SC16IS752 from stray signal noise on the board.



CONNECTORS

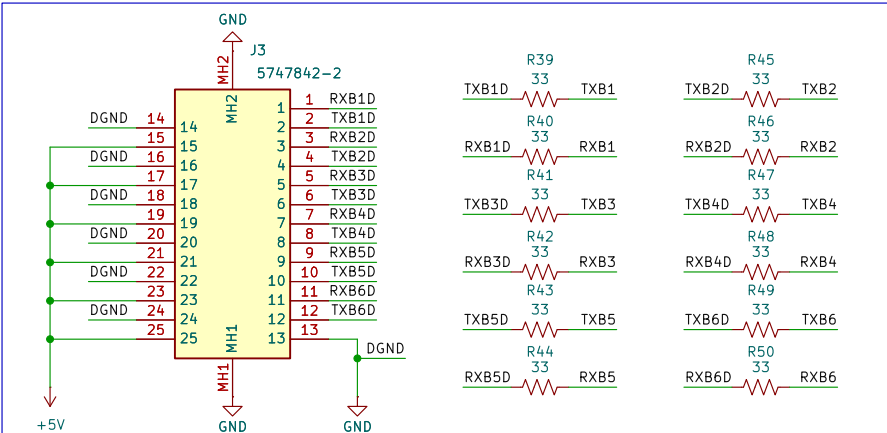
We need to connect 24 high frequency sensors to one PCB.

We'll need a wiring and connector solution that can take that kind of chaos.



Refer to page 8 for cable pinout.

This block handles sensor bank A, the RIGHT side connector. The 33 Ohm resistors are there to prevent signal reflections, increasing our signal quality.

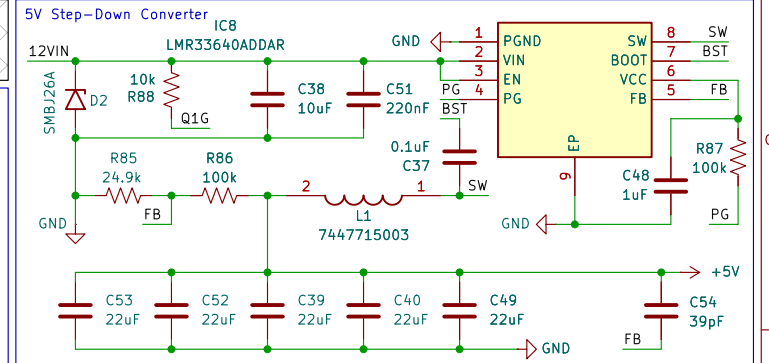
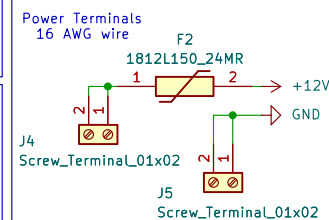
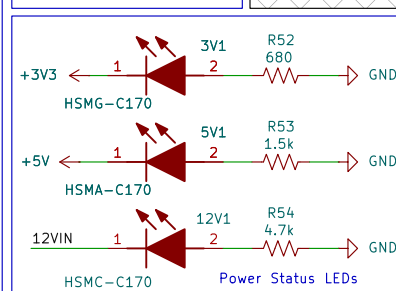
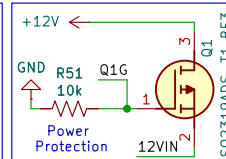


The other DB-25 connector for the TXB/RXB sides of the SC16IS752.

This block handles sensor bank B, the LEFT side connector.

The 33 Ohm resistors are there to prevent signal reflections, increasing our signal quality.

Refer to page 8 for cable pinout.



Max Current	Wire Gauge
2A	18 AWG
5A	16 AWG
10A	14 AWG
15A	12 AWG

NOTE:
Despite two terminals, the power terminals share the same island. This allows the terminals to "daisy-chain" to another device that requires up to 12V DC power. Be wary of voltage drop over cable length and no. of devices when doing this.
Calculate total current draw and refer to table for appropriate wire gauges. The included terminals fit 16 AWG wire. Assume this board generates 3.5A of current.

William Dinkins
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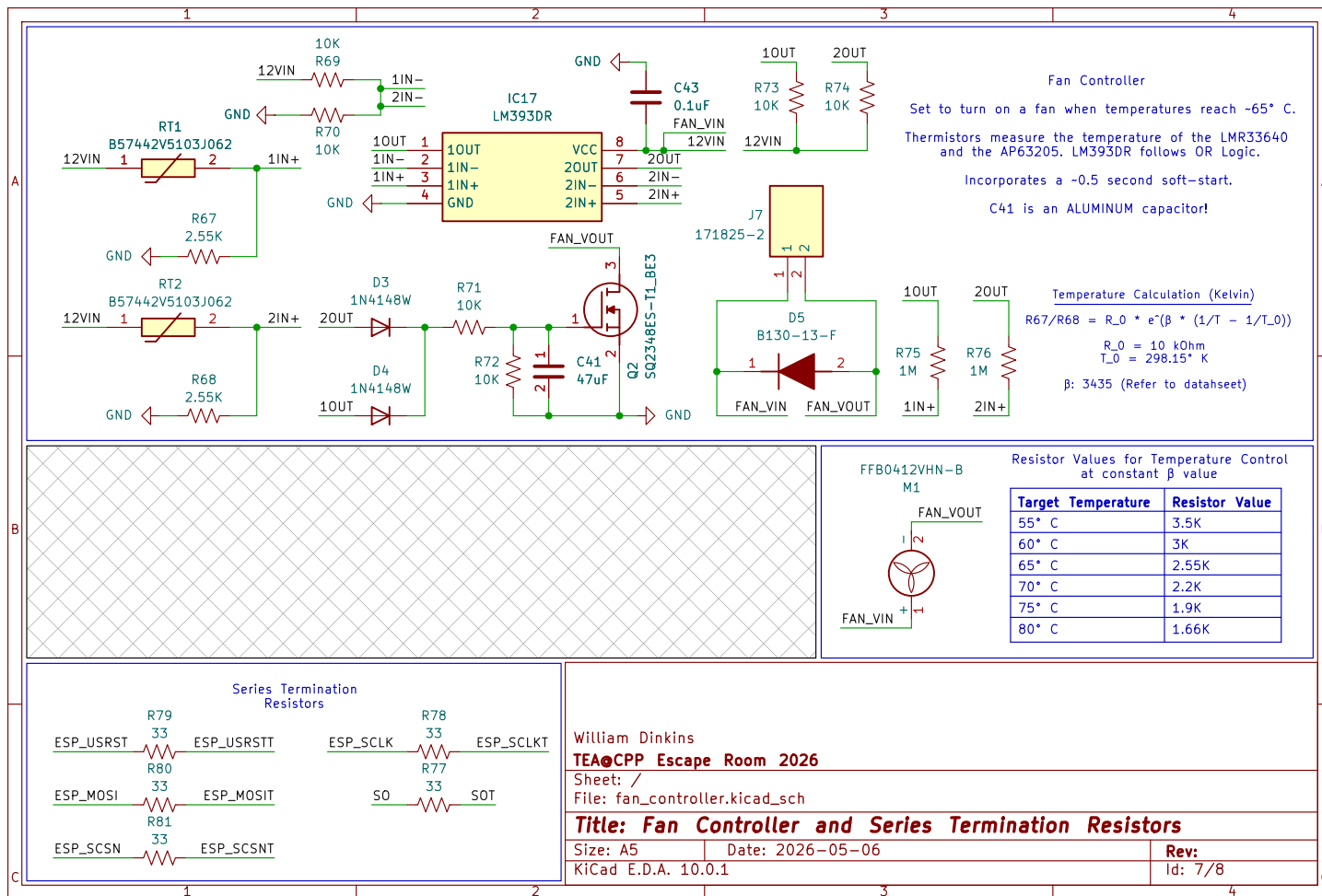
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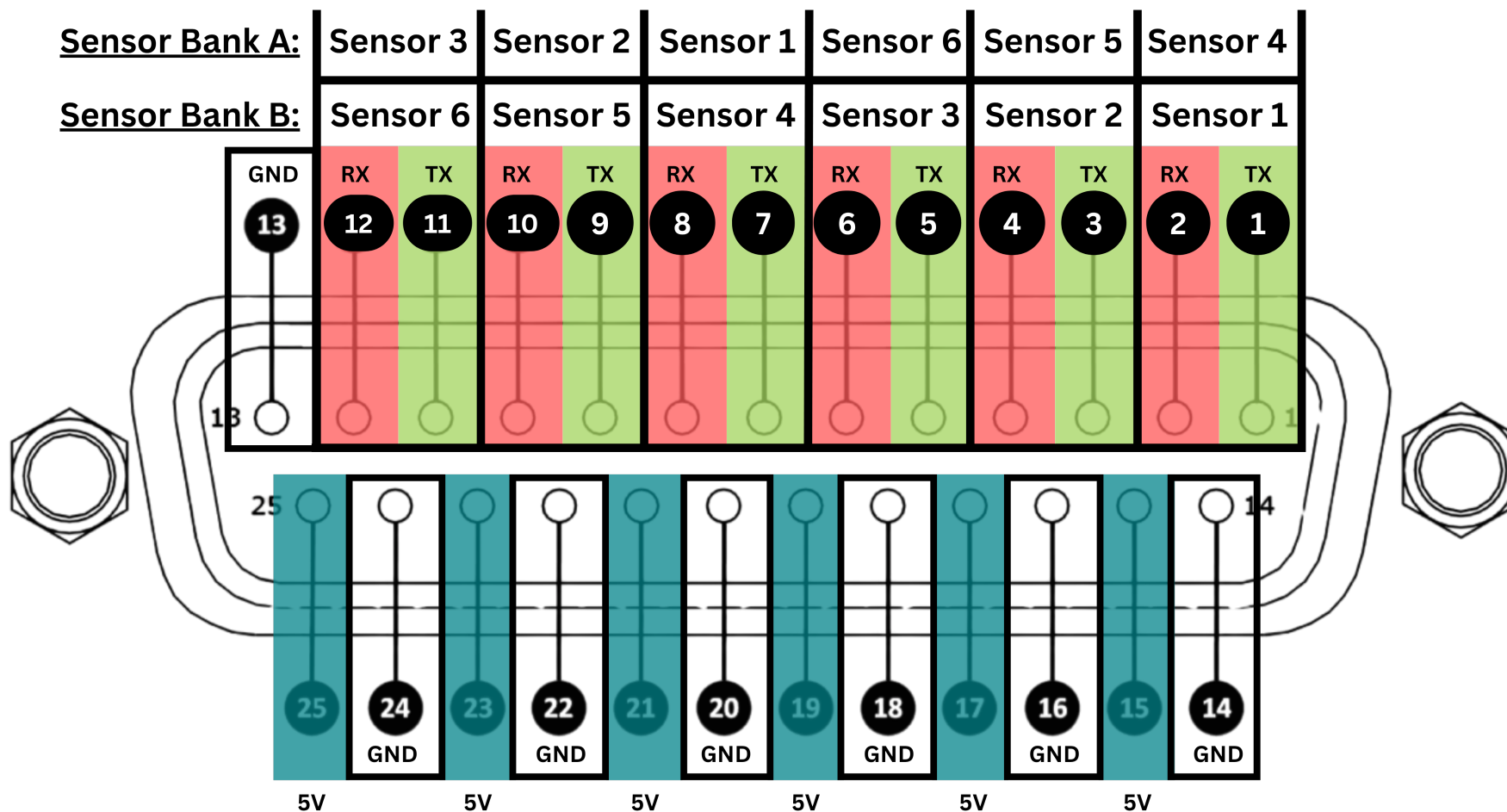
Title: Connectors, Timing, and Power Management

Size: A4 Date: 2026-04-26

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Rev: 1
Id: 6/8





This is the pinout for a custom female DB25 cable. This board requires a custom cable to connect to its sensors. Use this guide when making a new cable. Cable RX goes to Board TX, same for TX to RX.

Make sure to check all connections prior to connecting to power.

Incorrect wiring can cause permanent damage to sensors and/or board.

Use a multimeter set to continuity mode to ensure connections. Have a team member verify when finished.

Print this page out and have an installer sign here to verify connections:

Sheet: /
File: sensor_pinout.kicad_sch

Title: DB25 Pinout

Size: A4

Date:

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Id: 8/8